INCH-POUND

A-A-52091
March 14, 1991

COMMERCIAL ITEM DESCRIPTION

BLEACH, LAUNDRY, ORGANIC CHLORINE

The General Services Administration has authorized the use of this commercial item description in preference to O-B-420.

Abstract. This commercial item description covers one type of organic chlorine, dry bleach intended for direct addition to the laundry wheel in field and industrial laundry operations. The bleach is not intended to disinfect water, fruit, or vegetables.

1. SALIENT CHARACTERISTICS

1.1 Materials. The bleach shall be a white powder or bead containing an organic, chlorine liberating compound. The organic bleach shall be clean, free flowing, homogeneous, and shall show no evidence of separation. The bleach shall not be compounded using trichloroisocyanuric acids or mixtures containing trichloroisocyanuric acids, and shall contain no caustic sodas. In addition, the bleach shall contain a dusting agent; reactivity with carbonates shall result in slight turbidity. The bleach shall also meet the requirements specified in table I.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commander, U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-IRT, Natick, MA 01760-5019.

AMSC N/A

FSC 6850

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TABLE I. Physical and chemical requirements.

Physical and chemical requirement	Minimum	Maximum	Test method
Available chlorine (percent)	15.0	17.0	1/
Phosphates, as P_2O_5, (sodium tripolyphosphate, tetrasodium pyrophosphate or mixtures thereof) (percent)	14.0	20.0	D 820 2/
Carbonates, as CO_2 (percent)	----	0.2	D 460 2/
pH of an aqueous solution	7.8	9.0	E 70 2/
Insoluble matter content (percent)	----	1.5	D 820 2/
Moisture content (percent)	----	2.0	D 820 2/
Loss of available chlorine on heating (percent)	----	20.0	3/
Loss on ignition (percent)	18.0	----	D 501 2/
Particle size, pass through (percent)			
16 mesh sieve	90.0	----	D 502 2/
200 mesh sieve	----	1.0	D 502 2/
Stain removal (when tested for tea stain, Orange Pekoe Cut Black) (percent)	78.0	----	4/

1/ Weigh a 10 gram sample of bleach to the nearest hundredth of a gram into a clean, dry 500 milliliter Erlenmeyer flask containing a magnetic stirring bar. Add 200 milliliters of distilled water and 50 milliliters of alkaline potassium iodide solution (200 grams KI and 120 grams NAOH per liter). Stir for at least 5 minutes to dissolve the bleach, then slowly add 50 milliliters hydrochloric acid solution (1:1 of hydrochloric acid and distilled water) while continuing to stir. Loss of iodine vapors to the atmosphere will give erroneous results. Titrate immediately with sodium thiosulfate solution (1.0N) to a straw color. Add 5 milliliters of starch solution (1.0 percent) and continue titration slowly to the disappearance of blue color.

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$$\text{Percent available chlorine} = \frac{\text{mL. } Na_2S_2O_3 \text{ X normality of } Na_2S_2O_3 \text{ X } 0.03546 \text{ X } 100}{\text{Weight of sample (grams)}}$$

2/ ASTM test method.

3/ A small Wood's metal bath shall be prepared with a 250-mL metal beaker filled to approximatley three-fourths capacity with molten Wood's metal. This bath shall be heated either over a Bunsen burner or in an electric beaker heating mantle with rheostat. A thermometer (0° to 360°C) shall be inserted in the Wood's metal 1 inch from the side of the beaker so the top of the mercury bulb is 1 inch below the surface. A safety shield shall then be placed in front of this assembly. The analyst should wear protective clothing and complete eye and face protection. Into a 1- by 8-inch Pyrex test tube 5.0 grams of the unground product shall be placed. The bulb of a thermometer, 0° to 360°C, shall be immersed in this sample and the test tube shall be supported over the Wood's metal bath so that the top of the sample is just below the level of the Wood's metal. Heat shall be applied to the metal bath at such a rate that the temperature of the sample under test does not lag more than 10°C below the temperature of the bath. The temperature of the bleach shall be watched constantly. Visual indications of rapid reaction such as mild explosion, fire, or sudden rapid rise to a temperature greater than that of the bath shall be considered as evidence of thermal decomposition. The temperature taken at the point of decomposition is not the temperature reached after reaction has taken place, but is rather the last observed temperature before the reaction.

4/ Preparation of stained swatches.

a. Cloth. The cloth shall be preshrunk, plain weave, permanent finish cotton and shall contain approximately 54 ends and 45 picks per square inch and weigh approximately 5.1 ounces per square yard. The initial reflectance of the cloth shall be determined using a Model 610 Photovolt or equivalent, and a green tri-stimulus filter. Thirty-two, 3.5- by 3.5-inch, swatches shall be prepared.

b. Staining. Bring 1 liter of tap water to a boil. Insert five tea bags (Orange Pekoe Cut Black) in the boiling water and boil for 5 minutes, then remove bags. Insert 32 cloth swatches (3.5 by 3.5 inches) and continue to boil for 5 minutes. Remove swatches, wring out, and dry in a clothes dryer at 180° ± 10°F for 30 minutes. Rinse in cold water and repeat drying procedure.

NOTE: Stained swatches older than 2 weeks shall be discarded and fresh swatches prepared. Record the reflectance of the stained swatches taking average readings of both sides. Use swatches stacked four deep for the readings in order to get a background of similar reflectance.

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Stain removal.

a. Apparatus. A Launder-Ometer with pre-heating load table, or similar machine, shall be used, in which 1-pint glass jars are held with their bases toward a horizontal shaft and 2 inches from its center or rotation, the shaft having a speed of rotation of 40 to 45 r.p.m. The temperature of the bath shall be maintained at 49° ± 2°C.

b. Reagents.

Water. The hardness of the water shall not exceed 50 p.p.m. calculated as calcium carbonate ($CaCO_3$).

Detegent solution. A 0.4 percent solution of a synthetic detergent (bead form) conforming to type III of P-D-245 shall be used.

Bleach solution. A solution of laundry bleach shall be prepared in a 2-liter volumetric flask using sufficient ground and mixed dry bleach to give a concentration of 200 p.p.m. available chlorine (0.02 percent). Weigh bleach sample directly into the flask from a tared container of bleach and determine sample weight by difference. Prepare this solution immediately before use.

c. Procedure. Measure into each of five 1-pint glass jars, each containing ten 1/4-inch stainless steel balls, 150 mL of detergent solution and 150 mL of bleach solution both solutions at a temperature of 49°C. Place two stained swatches in each jar. Cap the jars and place in washing machine. The machine shall be operated for 20 minutes at temperature of 49° ± 2°C. After washing, the swatches shall be removed, rinsed thoroughly in water at a temperature of 45° - 50°C, squeezed by hand to remove excess water, smoothed flat, and air dried. Measure the reflectance of the bleach swatches, taking average readings on both sides. Use swatches stacked four deep for readings in order to get a background of similar refectance.

Calculation. Calculate the total percent stain removed as follows:

$$\text{Total percent stain removed} = \frac{B - S}{O - S} \times 100$$

B = Average swatch reflectance after bleaching
S = Average swatch reflectance of stained swatch
O = Initial reflectance of original unstained swatch

1.2 Label. Unless otherwise specified in the contract or purchase order, each shipping container shall be labeled in accordance with the Federal Hazardous Substances Act.

1.3 Material safety data sheets. A Material Safety Data Sheet shall be prepared in accordance with the requirements of FED-STD-313 (see 5.3).

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2. QUALITY ASSURANCE

2.1 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

2.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using coversion tables contained in the latest version of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met.

3. PACKAGING

3.1 Packing. Twenty-five, 50, or 125 pounds of bleach shall be packed in fiber drum conforming to type I, grade A, class 2 of PPP-D-723. Each fiber drum shall be fitted with a plastic bag having a minimum thickness of 0.002 inch. The plastic bag shall be closed by heat sealing, double-folding the bag material and then taping it to the bag body, knotting the bag closure, or by use of a plastic or plastic-coated closure tie.

3.2 Palletization. When specified (see 5.1), the fiber drums shall be palletized in accordance with load type XIIa, XIIb or XVII of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C or D, or film bonding means F or G.

3.3 Marking. In addition to any special markings required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129. In addition, each shipping container shall be printed with the following special marking:

STORE IN A COOL, DRY PLACE

4. NOTES

4.1 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. When palletization is required (see 4.2).

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4.2 Sources of documents.

4.2.1 Source of government documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

4.3 Material safety data sheets. Contracting Officers will identify those activities requiring copies of Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in Appendix B of FED-STD-313.

MILITARY INTERESTS:

Custodians

Army - GL
Air Force - 68

Review activities

Army - MD
DLA - GS

User activity

Army - AR

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
DCG
HHS - FDA
JUS - FPI
USDA - APH
VA - OSS

PREPARING ACTIVITY:

Army - GL

Project 6850-1073

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